Mastering PowerBeamer Interactive PDF Presentations

A Hands-On Guide to Tools, Inking, Shortcuts, and Media

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PowerBeamer Project

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PRESENTER NOTES (Slide 1):

- Welcome to the interactive PowerBeamer tutorial deck!
- Notice how this note displays in the presenter window without appearing on the audience screen.
- Press **L** to activate the laser pointer and practice guiding attention.
- Use the left/right arrow keys or spacebar to advance slides.

Step 1: Annotation & Inking Suite

Live Pen & Highlighter Tools

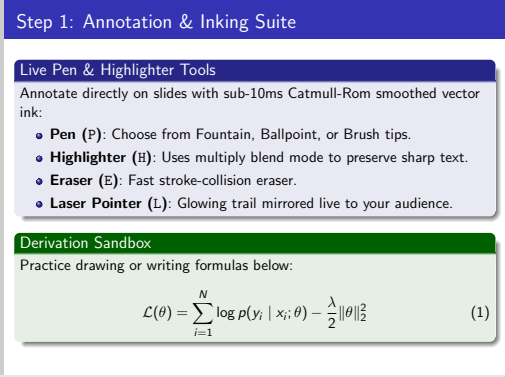
Annotate directly on slides with sub-10ms Catmull-Rom smoothed vector ink:

- **Pen (P)**: Choose from Fountain, Ballpoint, or Brush tips.
- **Highlighter (H)**: Uses multiply blend mode to preserve sharp text.
- **Eraser (E)**: Fast stroke-collision eraser.
- **Laser Pointer (L)**: Glowing trail mirrored live to your audience.

Derivation Sandbox

Practice drawing or writing formulas below:

$$\mathcal{L}(\theta) = \sum_{i=1}^{N} \log p(y_i \mid x_i; \theta) - \frac{\lambda}{2}\|\theta\|_2^2 \tag{1}$$



SPEAKER NOTES (Slide 2):

- Switch to the **Pen tool (P)** and circle the regularization term $\frac{\lambda}{2}\|\theta\|_2^2$.
- Try changing ink colors using the color palette in the top bar.
- Press **Cmd+Z** / **Ctrl+Z** to test multi-level undo.
- Press **N** or **+** to spawn a blank whiteboard whenever you need extra writing space!

Step 2: Embedded Audio Support

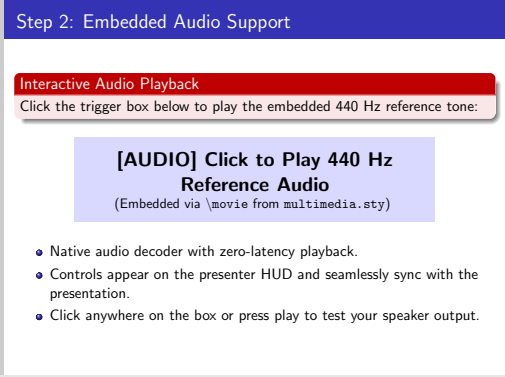
Interactive Audio Playback

Click the trigger box below to play the embedded 440 Hz reference tone:

[AUDIO] Click to Play 440 Hz Reference Audio

(Embedded via `\movie` from `multimedia.sty`)

- Native audio decoder with zero-latency playback.
- Controls appear on the presenter HUD and seamlessly sync with the presentation.
- Click anywhere on the box or press play to test your speaker output.



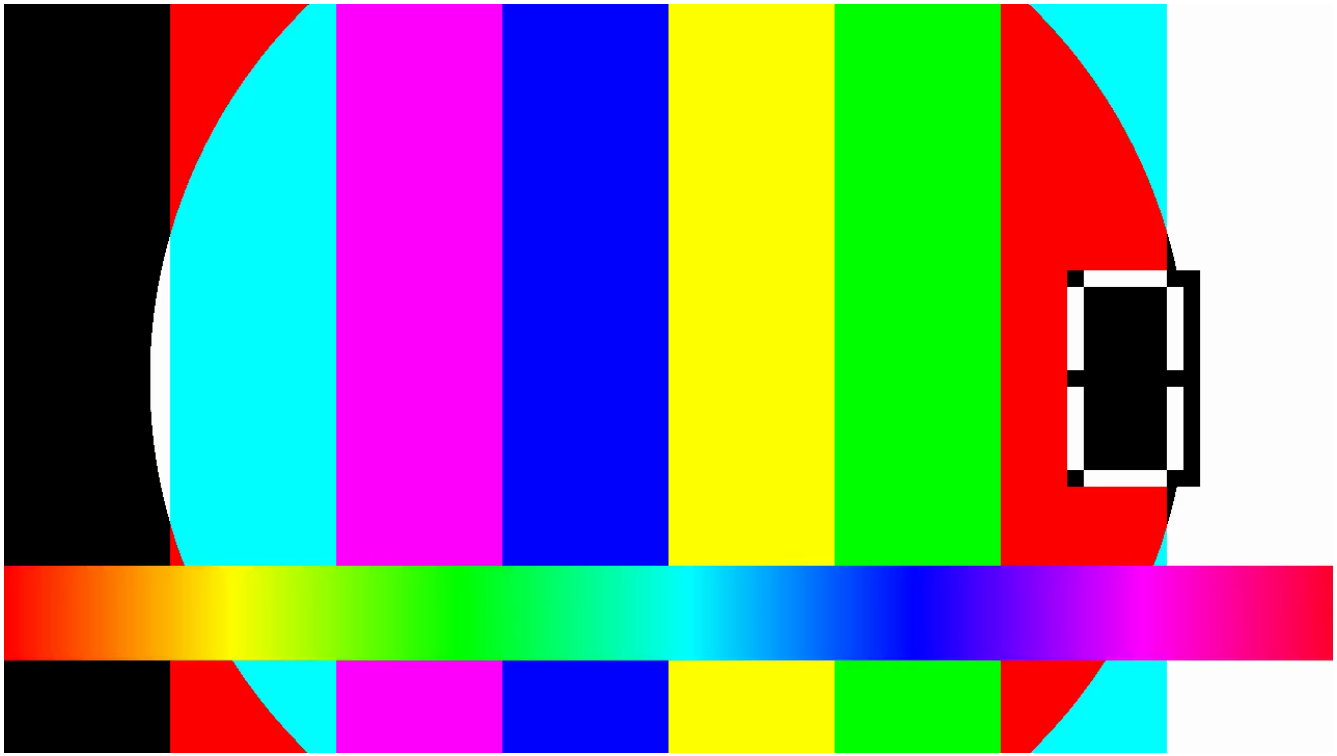
SPEAKER NOTES (Slide 3):

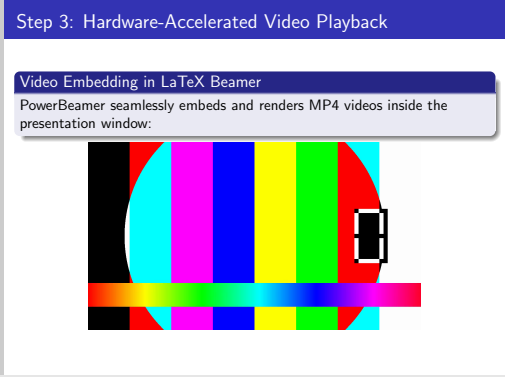
- Click the audio box above to verify media playback through your speakers.
- PowerBeamer extracts media clips defined with LaTeX `\movie` automatically.
- Check the audio volume and mute controls in the presenter controls toolbar.

Step 3: Hardware-Accelerated Video Playback

Video Embedding in LaTeX Beamer

PowerBeamer seamlessly embeds and renders MP4 videos inside the presentation window:





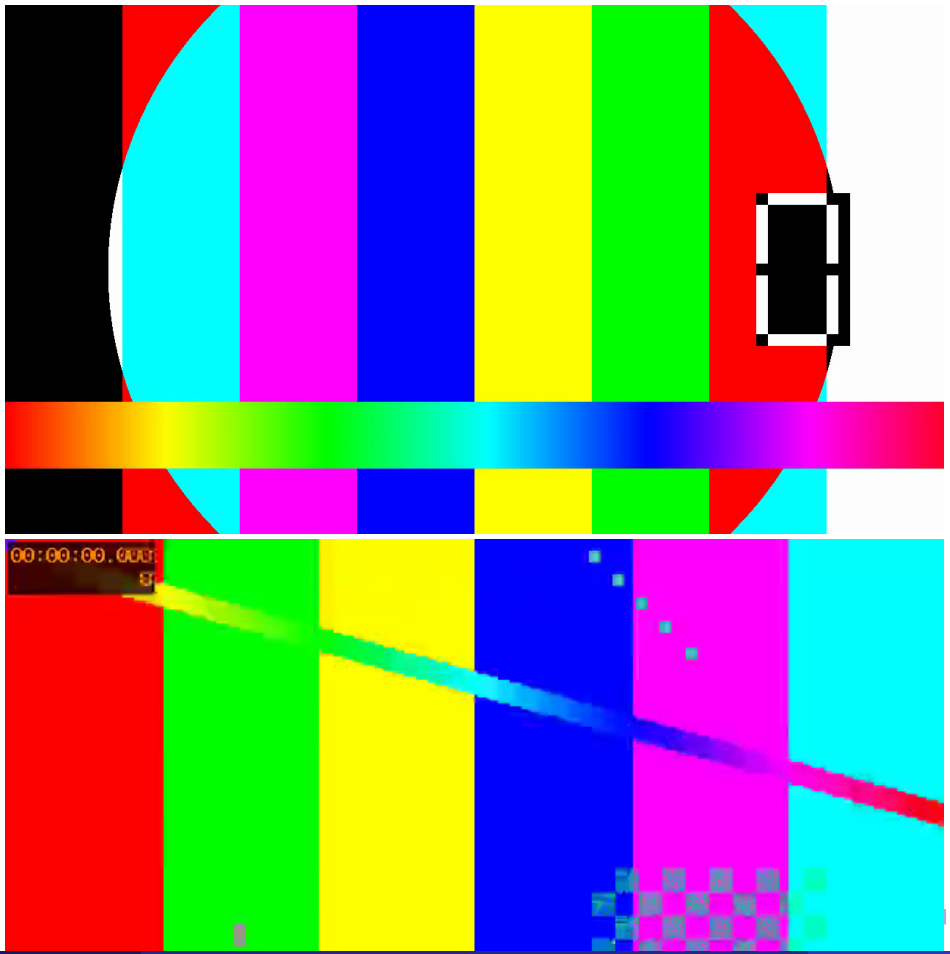
SPEAKER NOTES (Slide 4):

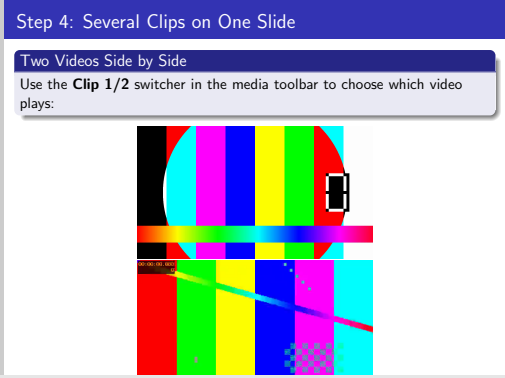
- Click on the video container to start playback.
- Note the hardware-accelerated decode with zero frame dropping.
- You can seek along the timeline scrub bar at the bottom.
- Audience and presenter screens stay perfectly synchronized.

Step 4: Several Clips on One Slide

Two Videos Side by Side

Use the **Clip 1/2** switcher in the media toolbar to choose which video plays:





SPEAKER NOTES (Slide 5):

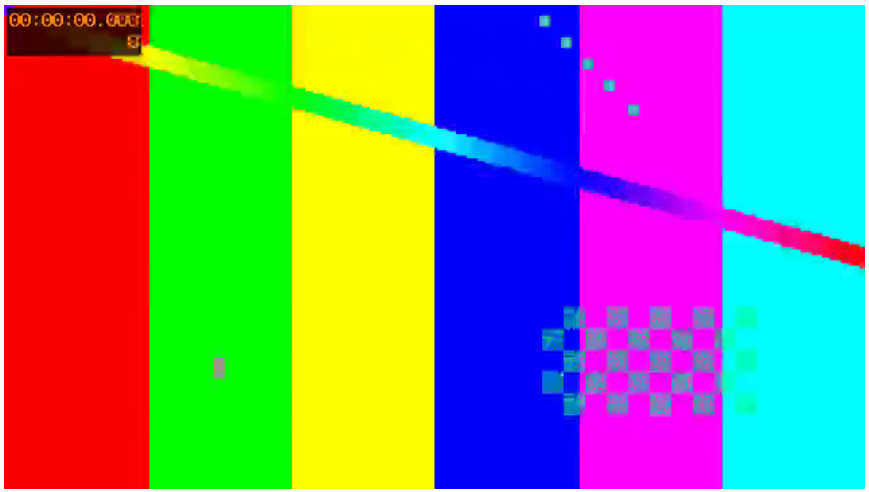
- The selected clip is outlined in red on the presenter screen.
- Switching clips stops the current one and resets the timeline.

Step 5: Audio Narration Next to a Video

Mixed Media

The audio clip comes first in the source, so it is selected by default:

[AUDIO] 440 Hz Narration Track





SPEAKER NOTES (Slide 6):

- Press play for the narration, then switch to Clip 2/2 for the video.

Step 6: Essential Shortcuts & Dual Display

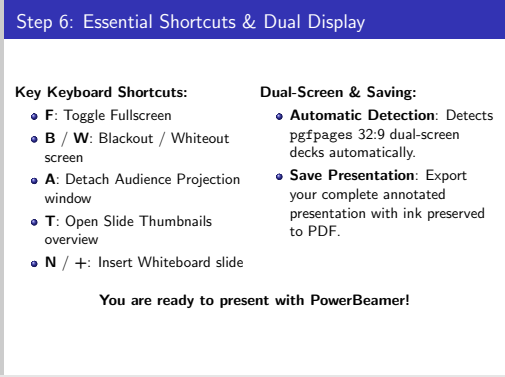
Key Keyboard Shortcuts:

- **F**: Toggle Fullscreen
- **B** / **W**: Blackout / Whiteout screen
- **A**: Detach Audience Projection window
- **T**: Open Slide Thumbnails overview
- **N** / **+**: Insert Whiteboard slide

Dual-Screen & Saving:

- **Automatic Detection**: Detects `pgfpages` 32:9 dual-screen decks automatically.
- **Save Presentation**: Export your complete annotated presentation with ink preserved to PDF.

You are ready to present with PowerBeamer!



SPEAKER NOTES (Slide 7):

- Wrap up the tutorial walkthrough.
- Press **T** to view the slide thumbnail grid and rearrange or navigate between slides.
- Test saving the annotated PDF from the top action menu.